

17.1 Optimal Adversarial Loss

We continue with the definition of Adversarial Risk defined in last class as:

$$L_N(h) = \mathbb{E}_{(\tilde{x}, y) \sim P^*} \left[\sup_{\tilde{x}_{\text{adv}} \in N(\tilde{x})} \ell((\tilde{x}_{\text{adv}}, y), h) \right].$$

Consider the neighbourhood function:

$$N(\cdot) = \{ \tilde{x} \text{ s.t. } \|\tilde{x} - \cdot\|_2 \leq \varepsilon \}.$$

For some classifier to be able to distinguish the 2 classes perfectly, the neighbourhoods of the points in opposite classes should not intersect. Conversely, if the neighbourhoods of 2 points in opposite classes intersect, there cannot exist a classifier which achieves an adversarial loss of 0.

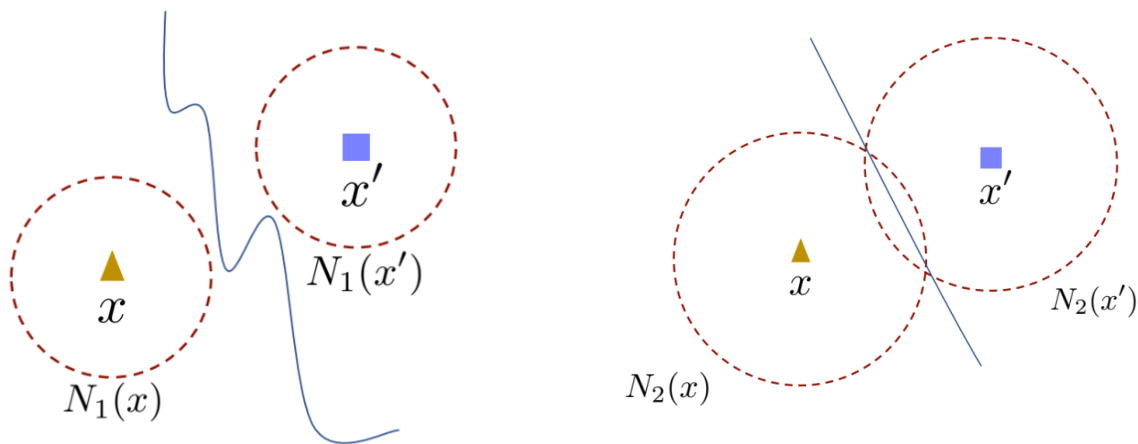


Figure 17.1. Examples of 2 neighbourhood functions

In the above example, there can exist an optimal classifier for $N_1(\cdot)$ but not for $N_2(\cdot)$.

Note: Counting the number of intersections at the distribution level will tell us how many samples *must* be misclassified.

17.1.1 Informal Theorem Statement

For given class-wise distributions and a specified adversary, we have

Lowest Robust Loss over a specific classifier family $\geq$ Lowest Robust Loss over all possible classifiers

where the lowest Robust Loss over all possible classifiers = a Simple function of an adversary-dependent distance between distributions. Moreover, there exists a classifier that achieves this optimal adversarial loss exactly.

Note that this is significant because the lowest Robust Loss over a specific classifier family is very hard to calculate in general, but the lowest Robust Loss over all possible classifiers is a solution of convex program for continuous and discrete distributions of interest.

Can we relate the optimal adversarial risk to the optimal transport cost under the adversarial cost function? [1] [2]

17.1.2 Formal Theorem Statement

Theorem 17.1. Consider the binary classification setting with $Y \in \{0, 1\}$, where $\vec{x}$ is drawn uniformly at random from P_0 and P_1 . The set of allowed hypotheses is $\mathcal{H} := \{h^A = 1_A(x \in A) : A \text{ is a closed set}\}$. Then, for the 0-1 loss function,

$$\forall h^A \in \mathcal{H}, \quad L_N(h^A) \geq \frac{1}{2} [1 - D_{N_\varepsilon}(P_0, P_1)].$$

Moreover, there exists some $h_N^* \in \mathcal{H}$ that achieves the lower bound.

17.1.3 Proof

Proof: Let $A \subseteq \mathcal{X}$ be a closed set where the classifier declares 1 on A and 0 elsewhere. Then,

$$\begin{aligned} \inf_{h^A \in \mathcal{H}} L_N(h^A) &= \inf_{A: \text{closed}} \frac{1}{2} (P_0(A^\varepsilon) + P_1((A^c)^\varepsilon)) \\ &= \frac{1}{2} \left(1 - \sup_{A: \text{closed}} [P_1(((A^c)^\varepsilon)^c) - P_0(A^\varepsilon)] \right) \quad \text{by Law of Total Probability} \end{aligned}$$

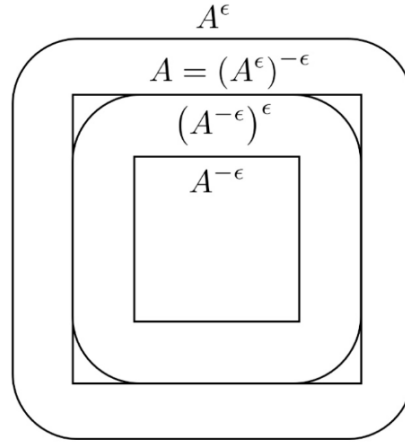
where, A^ε denotes the *expansion* of a set A , defined as

$$A^\varepsilon = \{\vec{x} \in \mathcal{X} : d(\vec{x}, \vec{x}') \leq \varepsilon \text{ for some } \vec{x}' \in A\}.$$

We also define

$$A^{-\varepsilon} = ((A^c)^\varepsilon)^c.$$

The following properties hold for Set Expansions and Complements:



- $((A^{-\epsilon})^{\epsilon})^{\epsilon} = (((A^{\epsilon})^{\epsilon})^{\epsilon})^{\epsilon} \subseteq A$
- $A \subseteq (A^{\epsilon})^{-\epsilon}$ and $(A^{\epsilon})^{-\epsilon} \subseteq A$. So, $(A^{\epsilon})^{-\epsilon} = A$

Theorem 17.2 (Strassen's Theorem). Let $\mathcal{X}$ be a Polish space, and let P, Q be probability measures on $\mathcal{X}$ with $\epsilon > 0$. Then

$$\begin{aligned} D_{N_{\frac{\epsilon}{2}}}(Q, P) &= \inf_{\pi \in \Pi(Q, P)} \mathbb{E}_{(\vec{x}, \vec{x}') \sim \pi} [1(d(\vec{x}, \vec{x}') > \epsilon)] \\ &= \inf_{\pi \in \Pi(Q, P)} \pi[\{(\vec{x}, \vec{x}') \mid d(\vec{x}, \vec{x}') > \epsilon\}] \\ &\geq \sup_{A: \text{closed}} [Q(A) - P(A^{\epsilon})], \end{aligned}$$

and some closed set A achieves the supremum.

Note: This is a special case of *Kantorovich duality*, which states that for lower semicontinuous cost functions $c(x, y)$, and for functions $\alpha(\cdot)$ and $\beta(\cdot)$ satisfying $\alpha(\vec{x}') - \beta(\vec{x}) \leq c(\vec{x}, \vec{x}') \quad \forall \vec{x}, \vec{x}'$,

$$\inf_{\pi \in \Pi(Q, P)} \mathbb{E}_{\pi}[c(x, x')] = \sup_{\alpha, \beta} (\mathbb{E}_P[\alpha(\vec{x}')] - \mathbb{E}_Q[\beta(\vec{x})]).$$

Coming back to the proof of the equivalence, it suffices to show that

$$\sup_{A: \text{closed}} [P_1(A^{-\epsilon}) - P_0(A^{\epsilon})] = \sup_{A: \text{closed}} [P_1(A) - P_0(A^{2\epsilon})].$$

Lemma 17.3. If A is closed, then both A^{ϵ} and $A^{-\epsilon}$ are closed.

Lemma 17.4. Let A be closed. Then $(A^{-\epsilon})^{\epsilon} \subseteq A$ and $A \subseteq (A^{\epsilon})^{-\epsilon}$.

Proof of Inequality Direction 1 (RHS $\geq$ LHS)

$$\begin{aligned} \sup_{A: \text{ closed}} [P_1(A) - P_0(A^{2\varepsilon})] &\geq \sup_{A: \text{ closed}} [P_1(A^{-\varepsilon}) - P_0((A^{-\varepsilon})^{2\varepsilon})] \\ &\geq \sup_{A: \text{ closed}} [P_1(A^{-\varepsilon}) - P_0(A^\varepsilon)], \end{aligned}$$

where the first step is a variable substitution which follows from 17.3 and last step follows from 17.4:

$$(A^{-\varepsilon})^{2\varepsilon} = ((A^{-\varepsilon})^\varepsilon)^\varepsilon \subseteq A^\varepsilon.$$

Thus, $P_0((A^{-\varepsilon})^{2\varepsilon}) \leq P_0(A^\varepsilon)$.

Proof of Inequality Direction 2 (LHS $\geq$ RHS)

$$\begin{aligned} \sup_{A: \text{ closed}} [P_1(A^{-\varepsilon}) - P_0(A^\varepsilon)] &\geq \sup_{A: \text{ closed}} [P_1((A^\varepsilon)^{-\varepsilon}) - P_0((A^\varepsilon)^\varepsilon)] \\ &\geq \sup_{A: \text{ closed}} [P_1(A) - P_0(A^{2\varepsilon})] \end{aligned}$$

where first step is a variable substitution, and second step follows from 17.4.

Now, by Strassen's theorem and the definition of $D_N^\varepsilon(P_0, P_1)$,

$$\begin{aligned} \inf_{h^A \in \mathcal{H}} L_N(h^A) &= \frac{1}{2} \left(1 - \sup_{A: \text{ closed}} [P_1(A^{-\varepsilon}) - P_0(A^\varepsilon)] \right) \\ &= \frac{1}{2} \left(1 - \sup_{A: \text{ closed}} [P_1(A) - P_0(A^{2\varepsilon})] \right) \\ &= \frac{1}{2} (1 - D_{N\varepsilon}(P_0, P_1)). \end{aligned}$$

□

Corollary 17.5 (Link to the Wasserstein Distance). *Under the same conditions as the preceding theorem, we have*

$$\inf_{h^A \in \mathcal{H}} L_N(h^A) \geq \frac{1}{2} \left[1 - \left(\frac{W_r(P_0, P_1)}{2\varepsilon} \right)^r \right],$$

where $W_r(P_0, P_1)$ denotes the r -th Wasserstein distance.

Proof: Let $r \geq 1$. For any $\pi \in \Pi(P_0, P_1)$,

$$\begin{aligned} \mathbb{E}_\pi [1\{d(\tilde{x}, \tilde{x}') > 2\varepsilon\}] &= \mathbb{P}_\pi [d(\tilde{x}, \tilde{x}') > 2\varepsilon] \\ &= \mathbb{P}_\pi [d^r(\tilde{x}, \tilde{x}') > (2\varepsilon)^r] \\ &\leq \frac{\mathbb{E}_\pi [d^r(\tilde{x}, \tilde{y})]}{(2\varepsilon)^r}. \end{aligned}$$

Therefore,

$$\inf_{\pi \in \Pi} \mathbb{E}_\pi [1\{d(\tilde{x}, \tilde{y}) > 2\varepsilon\}] \leq \frac{\inf_{\pi \in \Pi} \mathbb{E}_\pi [d^r(\tilde{x}, \tilde{x}')] }{(2\varepsilon)^r} = \frac{W_r(P_0, P_1)^r}{(2\varepsilon)^r}.$$

□

17.1.4 Empirical Results

We now empirically compare the optimal minimum adversarial loss (derived theoretically) with the robust loss achieved by trained classifiers under adversarial attacks.

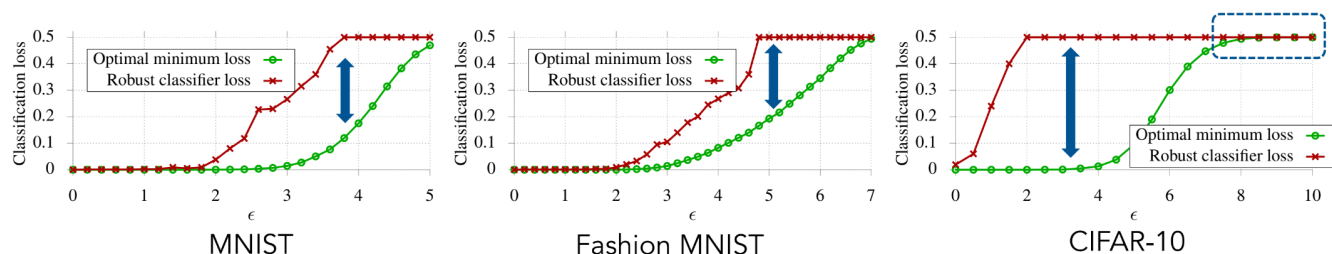
Experimental setup

- Train a robust classifier using Projected Gradient Descent (PGD) to maximize a proxy loss:

$$h_{\text{robust}} = \min_h \mathbb{E} \left[\max_{\|\delta\| \leq \epsilon} L_{\text{proxy}}((x + \delta, y), h) \right].$$

- Compute the loss on the training data using PGD attacks for different perturbation radii ϵ .
- Compute the optimal minimum loss using minimum matching on a bipartite graph, which corresponds to the optimal transport solution between class distributions.

Results



Observations and Takeaways

- There is a large gap between the optimal and learned (PGD-trained) robust losses, which increases with ϵ .
- For large ϵ , correct classification becomes impossible, as both the learned and optimal losses approach 0.5 (random guessing performance).

Bibliography

- [1] Arjun Nitin Bhagoji, Daniel Cullina, and Prateek Mittal. Lower bounds on adversarial robustness from optimal transport. *Advances in Neural Information Processing Systems*, 32, 2019.
- [2] Muni Sreenivas Pydi and Varun Jog. Adversarial risk via optimal transport and optimal couplings. In *International Conference on Machine Learning*, pages 7814–7823. PMLR, 2020.